

Arjun Giriraj Day 13

1. What is a network port?
Network port is the logical communication endpoint used by network application and service. Port number tells operating system which application should receive which traffic.
2. How many port numbers are available conceptually?
65535
3. What is a network protocol?
Network protocol is a set of communication rules like TCP, UDP, HTTP, HTTPS, etc.
4. What is TCP?
Transmission Control Protocol. It is connection-oriented, reliable-delivery, and use acknowledgments to confirm whether the traffic is received by the receiver.
5. What is UDP?
User Datagram Protocol. It is connectionless, it does not provide deliver guarantee. It is generally used where speed matters like video streaming or else.'
6. Explain two differences between TCP and UDP.
 - TCP is slow but connection oriented; UDP is faster but connectionless.
 - TCP used acknowledgement for cheking delivery; UDP does not guarantee deliver of traffic.
7. What is a listening port?
The port which is ready to have connection.
8. What is an ephemeral/client port?
Ports between 49152 and 65535 are used by client.
9. What is a socket?
A socket is the communication endpoint which contains IP address, protocol, port.
10. Which port is commonly used by SSH?
22
11. Which port is commonly used by HTTP?
80
12. Which port is commonly used by HTTPS?
443
13. Which port is used by DNS?
53
14. Which UDP ports are commonly used by DHCP client/server?
67/68

15. What is TCP port 445 commonly associated with?
SMB
16. What is port 3389 commonly associated with?
RDP
17. What does ss -tln display?
Listening ports.
18. Does an open port automatically mean a vulnerability? Explain.
No, open port does not automatically mean a vulnerability as firewalls manages open ports by blocking the inbound service connection which mismatch the source IP, destination IP, source Port, destination Port.
19. Why do firewalls use protocol and port information?
Firewalls uses protocol and port information to manage open ports by allowing known protocol and port and by blocking the inbound service connection that mismatches the source IP, destination IP, source port, and destination Port.
20. Why are ports and protocols important in cybersecurity?
Because Ports helps identify running services, network exposure, unexpected services, firewall rules, server configuration and much.

PRACTICAL EVIDENCE (your own VM/computer only):

ss -tln · ss -ltn · curl -I https://example.com

```
venom@VENOM:~$ ss -tln
```

Netid	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port
udp	UNCONN	0	0	127.0.0.54:53	0.0.0.0:*
udp	UNCONN	0	0	127.0.0.53%lo:53	0.0.0.0:*
udp	UNCONN	0	0	0.0.0.0:43512	0.0.0.0:*
udp	UNCONN	0	0	224.0.0.251:5353	0.0.0.0:*
udp	UNCONN	0	0	224.0.0.251:5353	0.0.0.0:*
udp	UNCONN	0	0	224.0.0.251:5353	0.0.0.0:*
udp	UNCONN	30080	0	0.0.0.0:5353	0.0.0.0:*
udp	UNCONN	0	0	0.0.0.0:5355	0.0.0.0:*
udp	UNCONN	0	0	*:1716	*:*
udp	UNCONN	0	0	*:54399	*:*
udp	UNCONN	8960	0	*:5353	*:*
udp	UNCONN	0	0	:::5355	:::*
tcp	LISTEN	0	4096	127.0.0.53%lo:53	0.0.0.0:*
tcp	LISTEN	0	4096	127.0.0.1:11434	0.0.0.0:*
tcp	LISTEN	0	4096	0.0.0.0:5355	0.0.0.0:*
tcp	LISTEN	0	4096	127.0.0.54:53	0.0.0.0:*
tcp	LISTEN	0	4096	:::5355	:::*
tcp	LISTEN	0	50	*:1716	*:*

```
venom@VENOM:~$ ss -ltn
```

State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port
LISTEN	0	4096	127.0.0.53%lo:53	0.0.0.0:*
LISTEN	0	4096	127.0.0.1:11434	0.0.0.0:*
LISTEN	0	4096	0.0.0.0:5355	0.0.0.0:*
LISTEN	0	4096	127.0.0.54:53	0.0.0.0:*
LISTEN	0	4096	:::5355	:::*
LISTEN	0	50	*:1716	*:*

```
venom@VENOM:~$ curl -i google.com
HTTP/1.1 301 Moved Permanently
Location: http://www.google.com/
Content-Type: text/html; charset=UTF-8
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-SePzle4USX4Q5N9kIjCpKQ' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http:;report-uri https://csp.withgoogle.com/csp/gws/other
Date: Mon, 21 Sep 2026 11:18:26 GMT
Expires: Wed, 21 Oct 2026 11:18:26 GMT
Cache-Control: public, max-age=2592000
Server: gws
Content-Length: 219
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN

<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
```